Image Super-Resolution via a Convolutional Neural Network (SRCNN)

UE24CS352A Machine Learning · Mini-Project Write-up
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1. Problem statement

Single-image super-resolution (SISR) recovers a high-resolution (HR) image from a low-resolution (LR) one. Interpolation (nearest, bilinear, bicubic) only smooths between known pixels and adds no information. We re-implement the SRCNN of Dong et al. (2015), as studied in the CS229 report by Garber, Grossman & Johnson-Yu, learn the LR→HR mapping from data, and compare it with the interpolation baselines using PSNR (higher is better).

2. Dataset

The reference work uses 100 DIV2K images (60 train / 20 val / 20 test), centre-cropped and resized to 224×224. **Our build environment had no internet access, so DIV2K could not be downloaded.** We therefore built a 100-image set with the same protocol (224×224 HR targets, 60/20/20) from random crops of the photographs bundled with scikit-image (train: astronaut, coffee, rocket, Hubble field, brick, gravel, grass, moon; val: cat, immunohistochemistry; test: retina, cameraman, cell). The split is by *source photo*, so val/test never share content with training. LR inputs are produced on the fly: HR → 112×112 (bilinear) → 224×224 (bicubic). Absolute PSNR values are therefore **not comparable** with the paper's DIV2K numbers; the repository includes `prepare_div2k.py` to rebuild the data from real DIV2K (not run by us).

3. Approach and implementation

Model. Three convolutions with 'same' padding: 9×9 (3→64) + ReLU for patch extraction, 1×1 (64→32) + ReLU for non-linear mapping, and 5×5 (32→3) for reconstruction (≈20k parameters). The input is the bicubic-upsampled LR image.
Implementation. Because PyTorch was unavailable, forward/backward passes, batch-norm, tanh, and the Adam optimiser are written in pure NumPy (im2col convolutions); a finite-difference test confirms every gradient (relative error < 1e-7). Training minimises MSE with Adam on random 48×48 patches (batch 16, flips, 320 patches/epoch, 40 epochs, one seed); evaluation uses full 224×224 images and the best-validation epoch. **Experiments** mirror the paper: three baselines; learning rates {1e-2, 1e-3, 1e-4}; tanh output; batch-norm after conv1/conv2; L2 regularisation {1e-2…1e-5}. As our own refinement we add a **residual SRCNN** (output = network(x) + bicubic input, so the net only learns the missing detail).

4. Results

Experiment	Max val PSNR (dB)	Test MSE	Test PSNR (dB)
Basic SRCNN (lr 0.001)	30.943	0.000538	36.567
+ Regularization (wd 1e-4)	30.881	0.000585	36.019
+ Tanh	28.717	0.000848	33.027
+ Batch Norm	27.197	0.001725	27.733
Residual SRCNN (refinement)	31.785	0.000432	37.848
Bicubic	29.866	0.000734	36.857
Bilinear	28.901	0.000937	35.351
Nearest	28.833	0.000946	34.432

Table 1: validation/test results (lr 0.001; MSE and PSNR on [0,1] images). Bilinear/nearest baselines upsample the 112×112 image directly. Highlighted: best model.

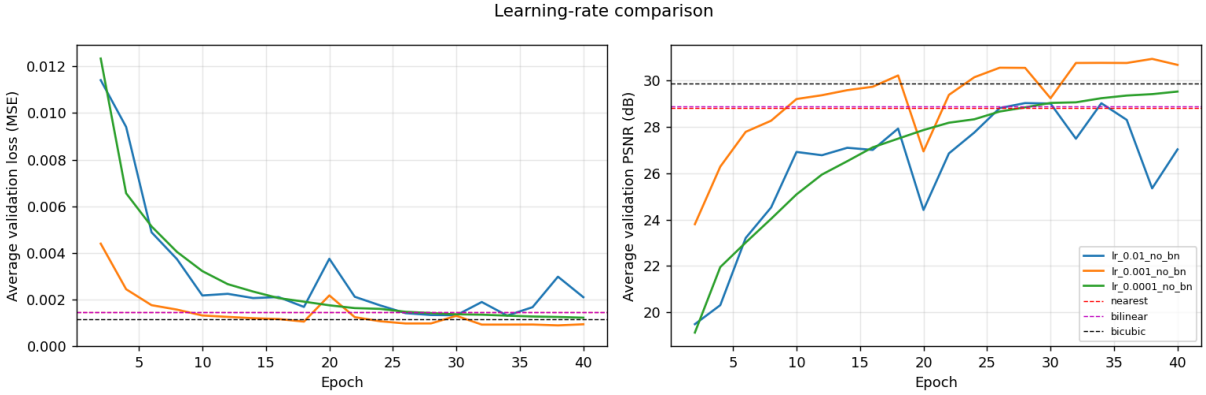


Figure 1: validation loss / PSNR vs epoch for three learning rates; dashed lines are interpolation baselines.

5. Conclusions

• **Learning rate 0.001 is best** (val 30.94 dB; 1e-2 is unstable at 29.03 dB, 1e-4 under-trains in 40 epochs at 29.53 dB), as in the paper.
• **Tanh and batch-norm hurt** (val 28.72 and 27.20 dB, vs 30.94 dB); batch-norm training curves are far noisier, as the paper reports. **L2 regularisation did not help**: 1e-5 matches the basic model (36.57 dB test) and larger factors degrade (1e-2: 29.77 dB).
• **Basic SRCNN beats bicubic on validation by 1.1 dB, but not on our test set** (36.57 vs 36.86 dB; better on only 9/20 images). Our test photos (retina, cell) are very smooth, and a plain SRCNN has to regenerate every pixel, introducing small colour/brightness shifts (Fig. 2, rows 1-2) that cost more PSNR than the sharpening gains. We state this openly rather than hide it.
• **Residual SRCNN fixes this**: val 31.79 dB and test 37.85 dB, i.e. +1.9 dB val and +1.0 dB test over bicubic, better on 16/20 test images. Because the refinement was added after seeing the basic-model test result, it should be read as a follow-up improvement, not a pre-planned test.

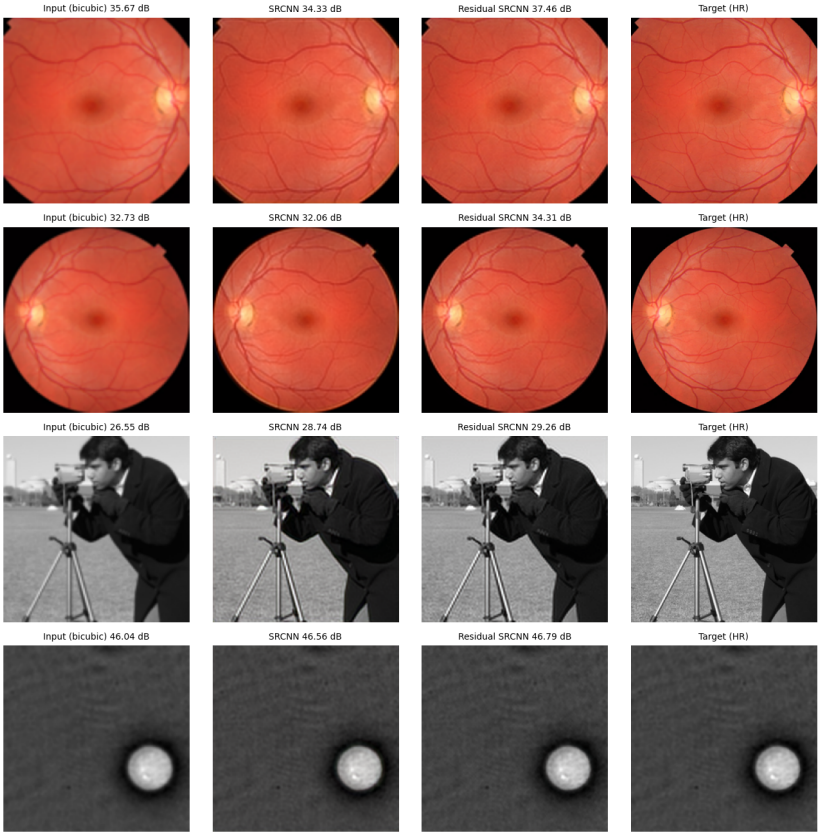


Figure 2: test examples — bicubic input, SRCNN, residual SRCNN, target.

Limitations / future work. Single seed, only 100 small images, 20 test images from 3 source photos, CPU-only 40-epoch training, so differences of a few tenths of a dB are not statistically established. Next steps: real DIV2K, multiple seeds, deeper models (VDSR/EDSR), and GAN-based losses.

Code: private GitHub repository shared with faculty/TAs (README explains setup and every command). **Reference:** C. Dong et al., “Image Super-Resolution Using Deep Convolutional Networks”, IEEE TPAMI 38(2), 2016.